

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

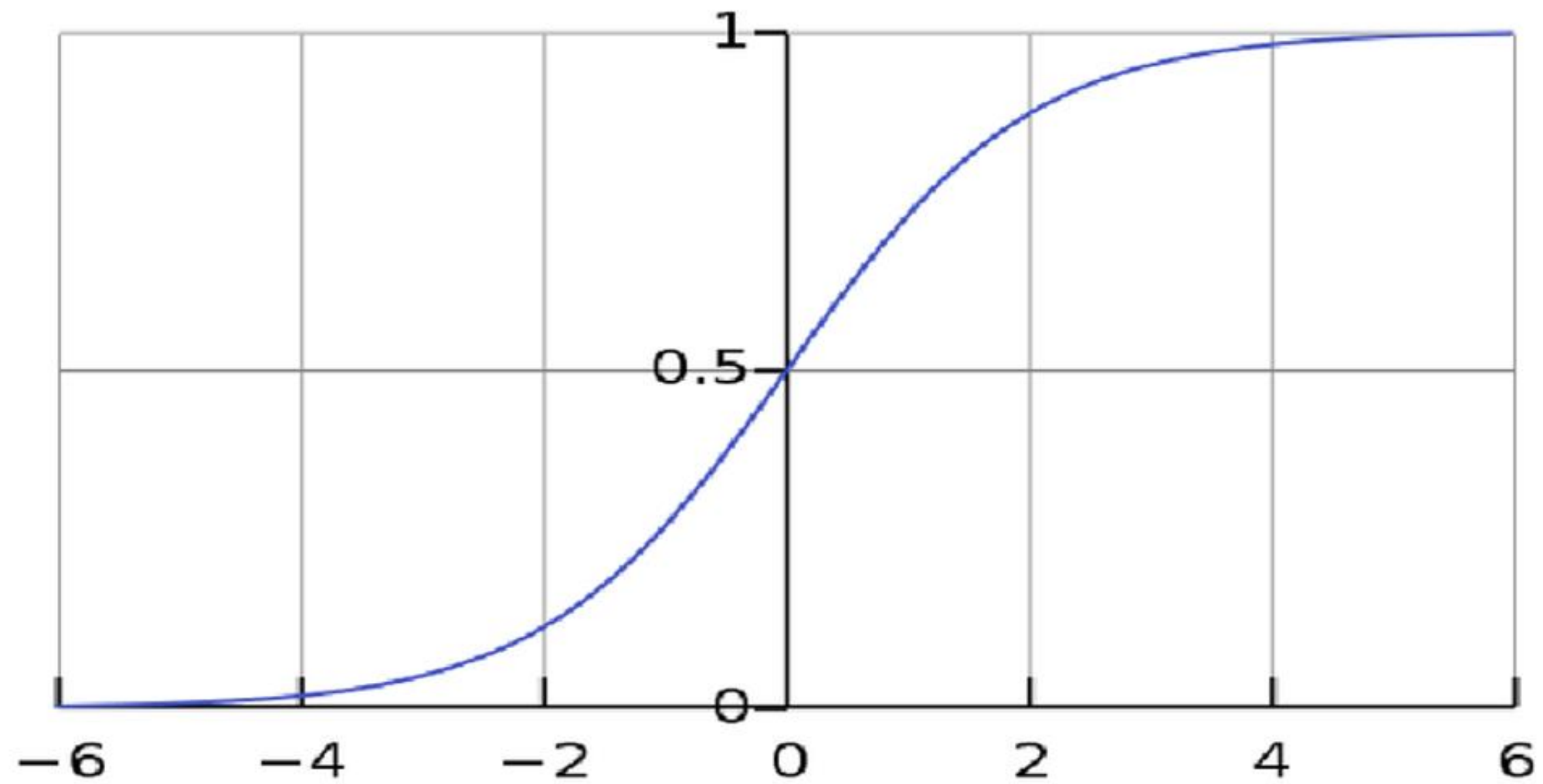
CSCE 633: Machine Learning

Lecture 2-3: Measures of Model Performance
Creating Operating Characteristic Curves

Measure of Model Performance: Topics

- Linear algebra review
- Supervised model introduction
- Metrics of performance
- Be able to create receiver operating characteristic curves
- kNN – a simple supervised learning model

Decision Threshold (1)



Decision Threshold (2)

- If we pick a decision threshold of $p(x) > 0.5$ what happens?

Predicted	Ground Truth
2%	0
3%	0
5%	1
1%	0
15%	0
25%	1
24%	1
13%	0
8%	0
12%	1

Decision Threshold (3)

↓

→

Predicted	TRUE	50	25	24	15	13	12	8	5	3	2	1
1	0	0	0	0	0	0	0	0	0	0	0	1
2	0	0	0	0	0	0	0	0	0	0	1	1
3	0	0	0	0	0	0	0	0	0	1	1	1
5	1	0	0	0	0	0	0	0	1	1	1	1
8	0	0	0	0	0	0	0	1	1	1	1	1
12	1	0	0	0	0	0	1	1	1	1	1	1
13	0	0	0	0	0	1	1	1	1	1	1	1
15	0	0	0	0	1	1	1	1	1	1	1	1
24	1	0	0	1	1	1	1	1	1	1	1	1
25	1	0	1	1	1	1	1	1	1	1	1	1

↑

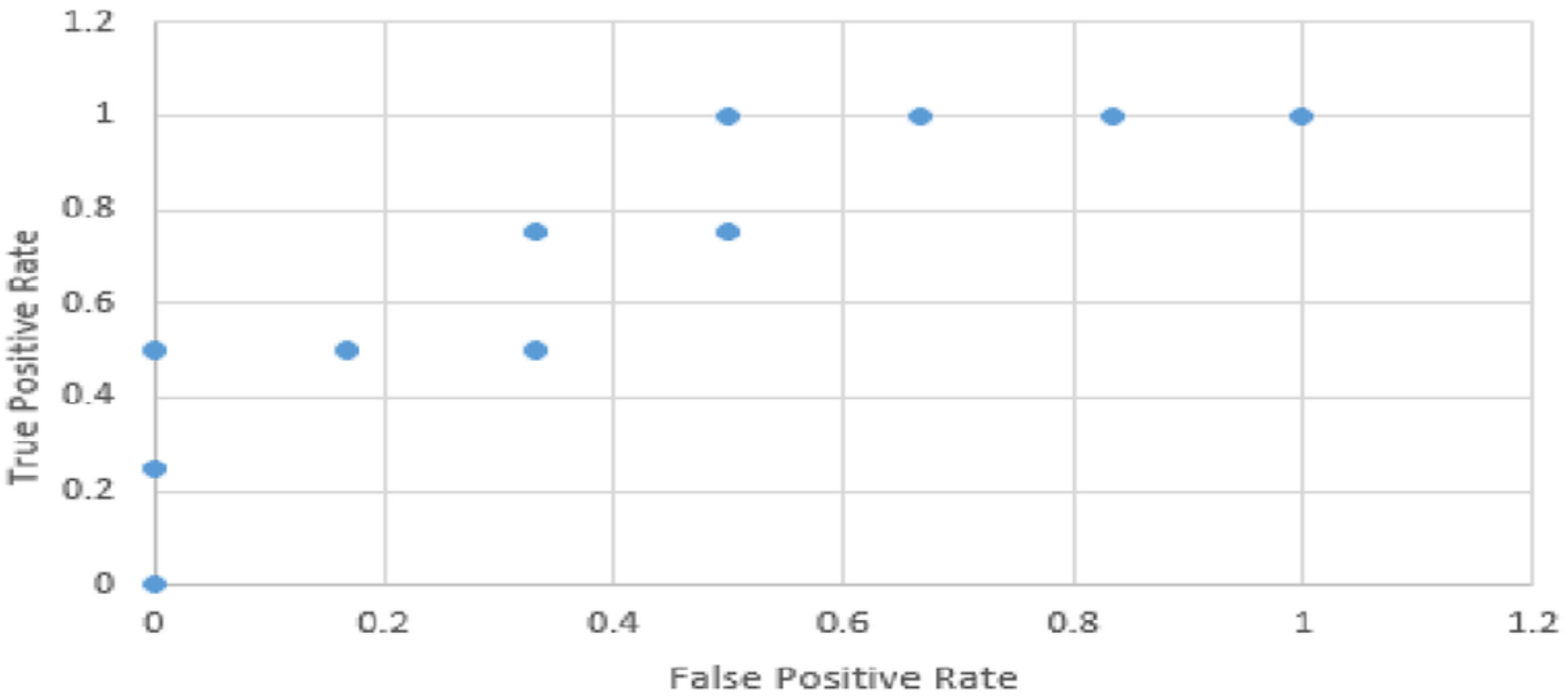
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Decision Threshold (4)

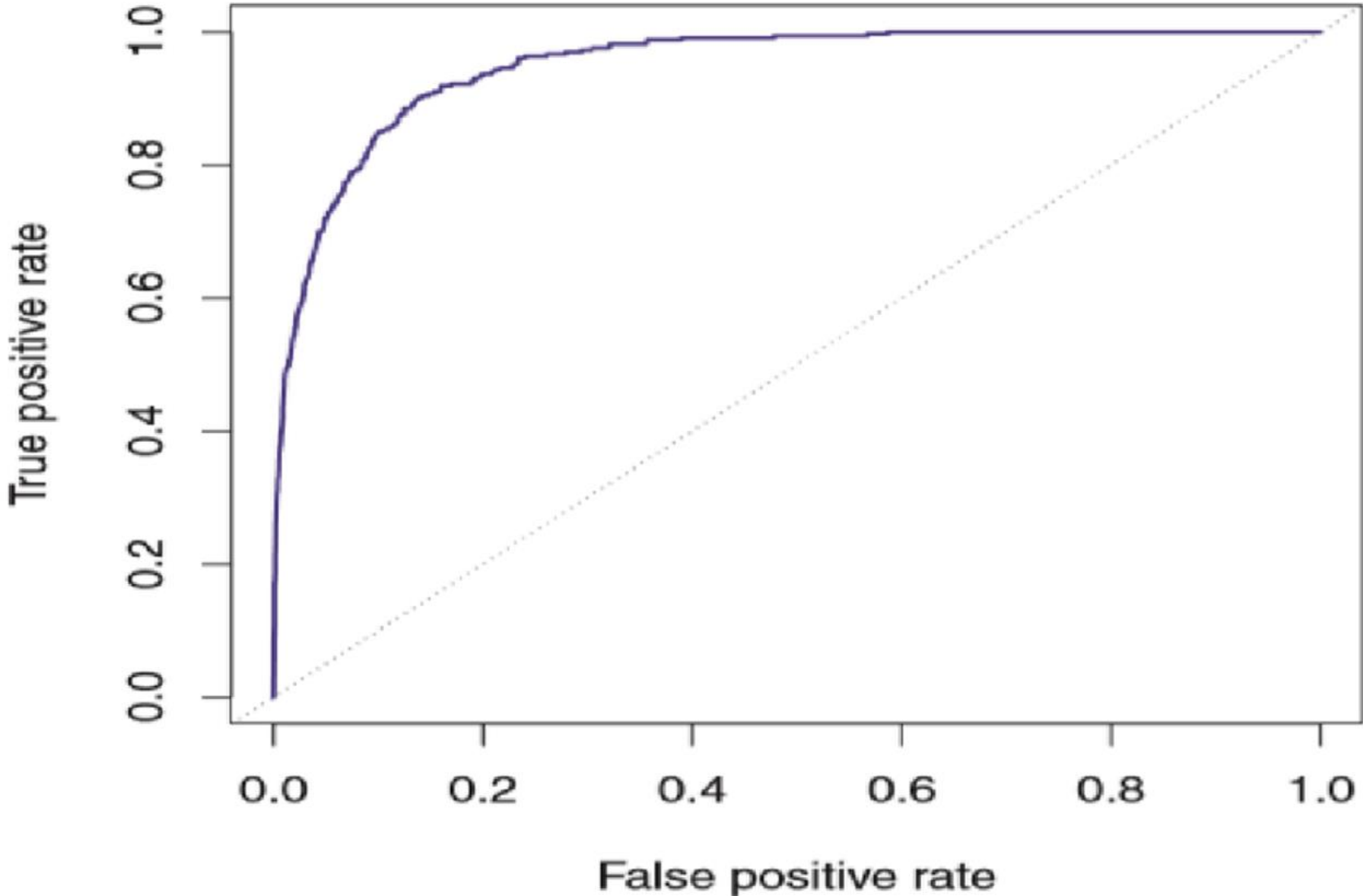
	50	25	24	15	13	12	8	5	3	2	1
TP	0	1	2	2	2	3	3	4	4	4	4
FP	0	0	0	1	2	2	3	3	4	5	6
TN	6	6	6	5	4	4	3	3	2	1	0
FN	4	3	2	2	2	1	1	0	0	0	0
TPR	0	0.25	0.5	0.5	0.5	0.75	0.75	1	1	1	1
FPR	0	0	0	0.166667	0.333333	0.333333	0.5	0.5	0.666667	0.833333	1

Decision Threshold (5)

	50	25	24	15	13	12	8	5	3	2	1
TP	0	1	2	2	2	3	3	4	4	4	4
FP	0	0	0	1	2	2	3	3	4	5	6
TN	6	6	6	5	4	4	3	3	2	1	0
FN	4	3	2	2	2	1	1	0	0	0	0
TPR	0	0.25	0.5	0.5	0.5	0.75	0.75	1	1	1	1
FPR	0	0	0	0.166667	0.333333	0.333333	0.5	0.5	0.666667	0.833333	1



Receiver Operating Characteristic Curve



AUC-ROC Curve (1)

What is AUC-ROC curve

- Area Under the Curve (AUC): the measure of separability
- Receiver Characteristic Operator (ROC): a probability curve
- AUC-ROC curve is a performance measurement for classification problem at various threshold settings
- It tells how much a model is capable of distinguishing between classes
- By predicting probability instead of label, can generate ROC Curve
- Can choose threshold on ROC curve that optimizes some threshold-specific measurement.
- Can also plot Precision-Recall Curve

AUC-ROC Curve (2)

- True Positive Rate (TPR) / Recall / Sensitivity:

$$TPR = \frac{TP}{TP + FN}$$

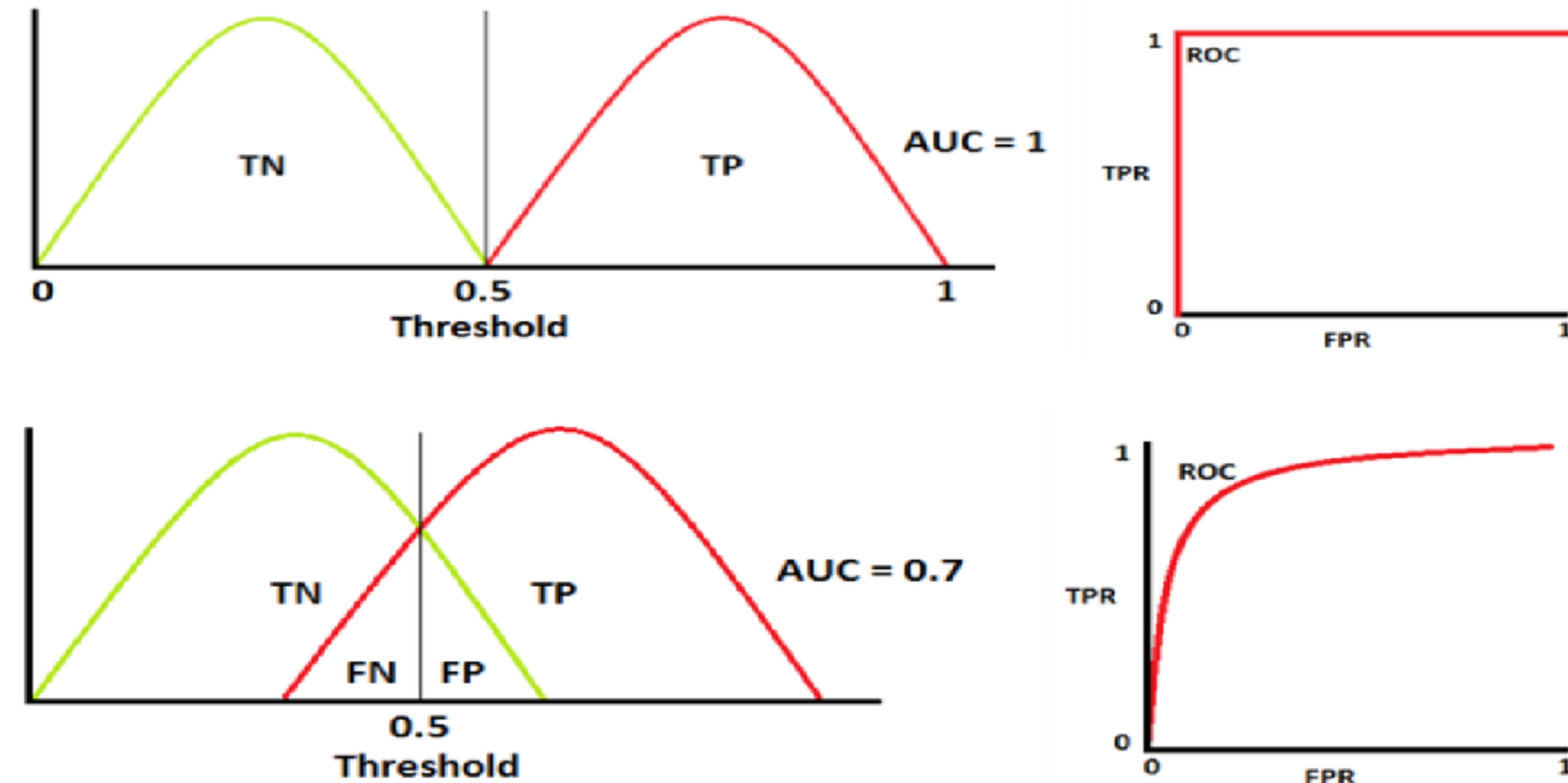
- Specificity:

$$Specificity = \frac{TN}{TN + FP}$$

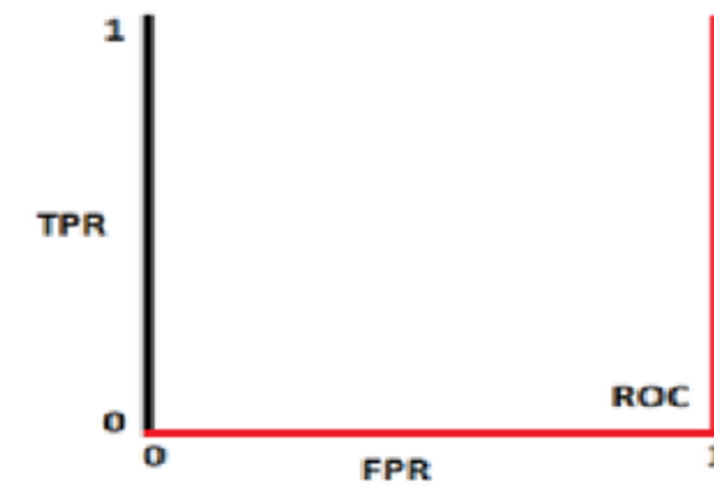
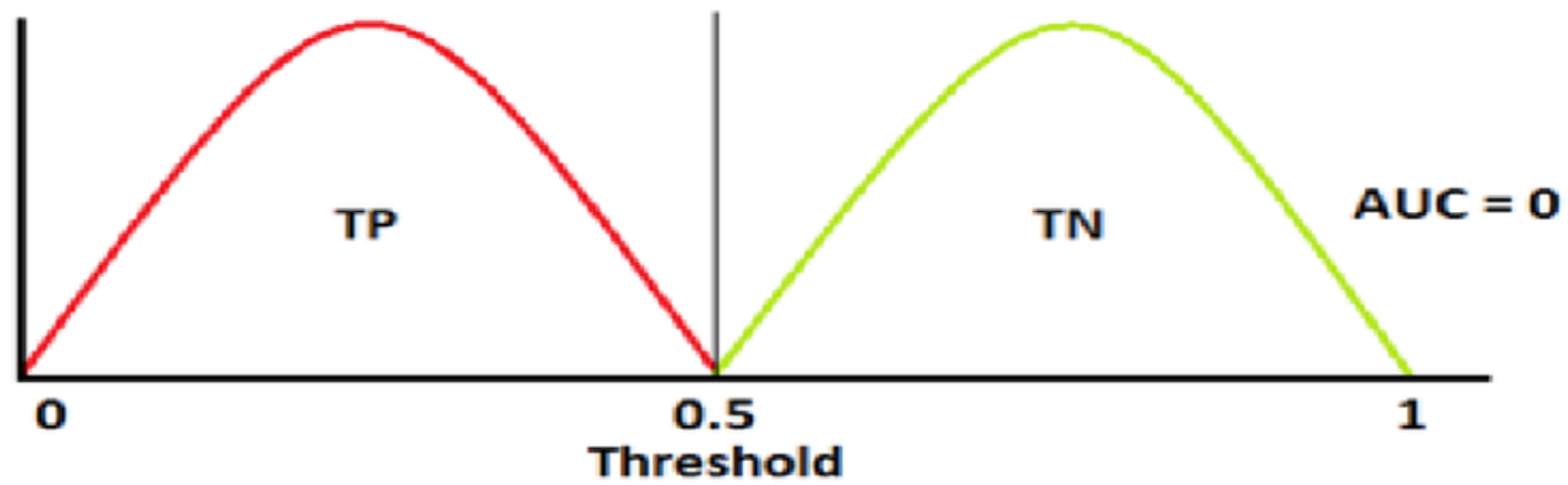
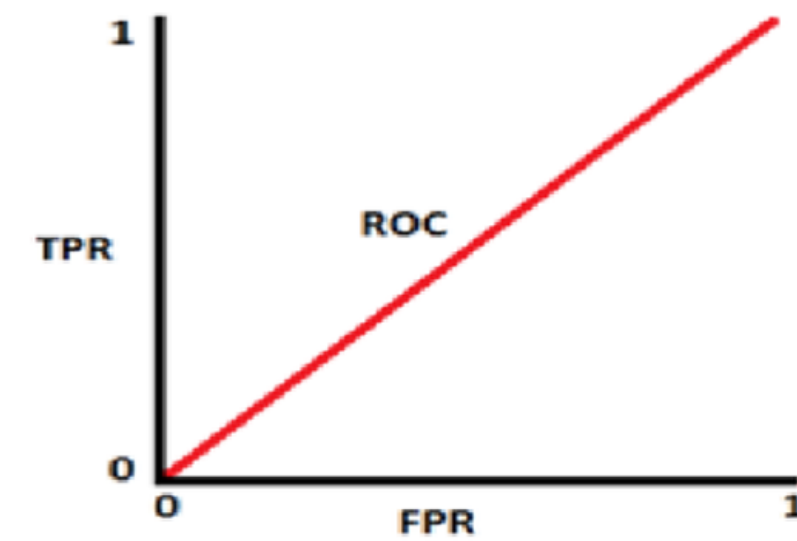
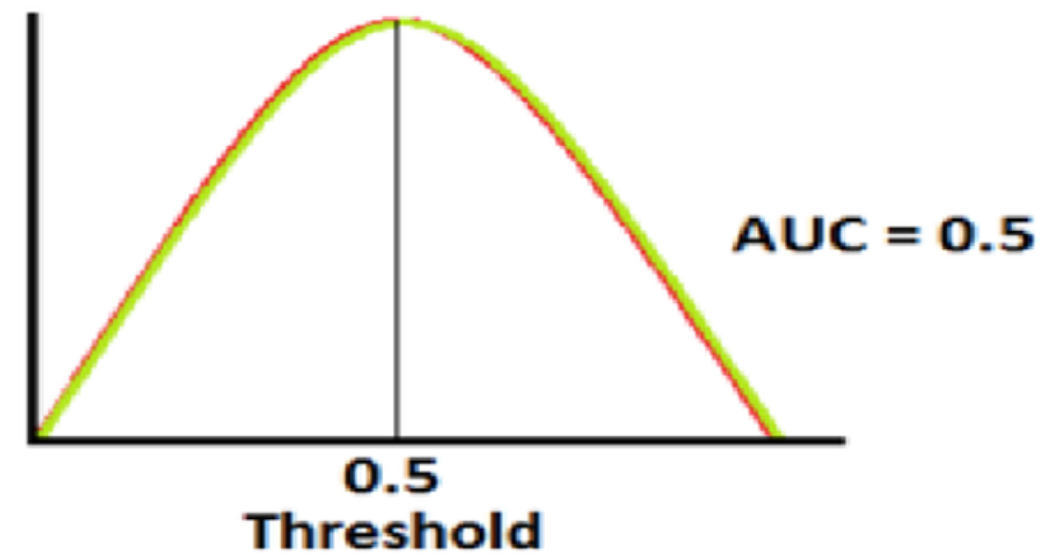
- False Positive Rate (FPR):

$$FPR = 1 - Specificity = \frac{FP}{TN + FP}$$

AUC-ROC Curve (3)



AUC-ROC Curve (4)



AUC-ROC Curve (5)

- Practice 3: create AUC-ROC curve from the given samples below:

Predicted	Ground Truth
13%	1
8%	0
4%	0
45%	1
75%	1
64%	1
23%	0
18%	0
32%	1
55%	0

AUC-ROC Curve (6)

	80	70	60	50	40	30	20	10	5
TP	0	1	2	2	3	4	4	5	5
FP	0	0	0	1	1	1	2	3	4
TN	5	5	5	5	4	4	3	2	1
FN	5	4	3	2	2	1	1	0	0
TPR	0	0.2	0.4	0.5	0.6	0.8	0.8	1	1
FPR	0	0	0	0.1666667	0.2	0.2	0.4	0.6	0.8

